॥श्रीगणेशाय नमः॥

Decision Trees and Ensembling

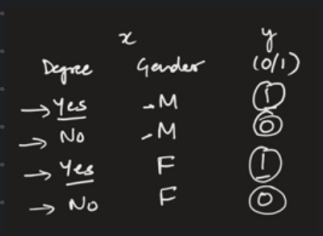
Decision trees

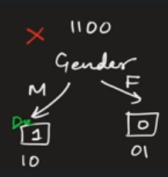
- Fundamental ML tools for both classification and regression tasks.

Structure of decision trees

- Nodes: Each node represents a feature in the dataset.
  Decision nodes: Nodes that split into branches based on features.
  Leaf nodes: Terminal nodes that provide the final output. Each leaf corresponds to a class label (in classification) or a value (in regression).

- Splitting: Dividing a node into 2/more sub-nodes based on a certain criteria. A good split will result in child nodes that are more homogeneous than the parent node.







- Depth of the tree: longest path from the root to a leaf. The depth affects the performance. Deeper trees may capture more detail but are also more prone to overfitting. If the tree isn't allowed to grow sufficiently, it may underfit.

Splitting criteria

Gini index: It measures the impurity in the dataset.

$$\text{Gini}(D) = 1 - \sum_{i=1}^{n} P_i^2$$

D → Dataset
K → # of classes
P_i → proportion of instances in class i

Consider a dataset with following class distribution:-

Class	# instances
A	10
B	5
C	5

Calculate the Gini index for this dataset.

$P_A = \frac{10}{20} = \frac{1}{2}$ $P_B = \frac{5}{20} = \frac{1}{4}$ $P_C = \frac{5}{20} = \frac{1}{4}$

$$\text{Gini}(D) = 1 - \left(\left(\frac{1}{2}\right)^2 + \left(\frac{1}{4}\right)^2 + \left(\frac{1}{4}\right)^2\right)$$

$$= 1 - \left(\frac{1}{4} + \frac{1}{16} + \frac{1}{16}\right) = 1 - \left(\frac{1}{4} + \frac{1}{8}\right) = 1 - \frac{12}{32} = 1 - \frac{3}{8} = \frac{5}{8} = 0.625$$

Gini index ranges from 0 to 1. [0, 1)
0 → perfect purity (all instances belong to a single class).
1 → max impurity (instances are divided among classes).

All instances belong to a single class → Gini ↓
All instances are equally divided among all classes → Gini ↑

Q. The maximum Gini index for binary classes is ___.

→ Gini index is maximized if there are equal # of examples in all classes.
2 classes → each with $\frac{1}{2}$ of the elements.

$$\text{Gini index} = 1 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 = \frac{1}{2}$$

Gini index $= 1 - P_A^2 - (1 - P_A)^2$ ← can differentiate and set to zero to get $P_A = 1/2$

(For 3 classes:

$$\text{Gini}_{max} = 1 - \left(\frac{1}{3}\right)^2 \cdot 3 = 1 - \frac{1}{3} = \frac{2}{3}$$

For 4 classes: $\frac{3}{4}$

For n classes: $1 - \left(\frac{1}{n}\right)^2 \cdot n = 1 - \frac{1}{n} = \frac{n-1}{n}$

For $\frac{n-1}{n} = 0$

$n = n - 1$

$n \to \infty$

For Gini index = 1, we need ∞ classes)

Entropy:
- It quantifies the randomness in the information being processed.
- It is a measure of uncertainty or impurity in a dataset.

$$H(D) = -\sum_{i=1}^{k} P_i \log_2 P_i$$

Class	# instances
A	10.
B	5.
C	5.

$P_A \cdot \log P_A = \frac{10}{20} \lg\left(\frac{10}{20}\right) = \frac{1}{2}\lg\left(\frac{1}{2}\right)$

$P_B \log P_B = \frac{5}{20} \log \frac{5}{20} = \frac{1}{4}\lg\left(\frac{1}{2}\right)^2 = \frac{1}{2}\lg\left(\frac{1}{2}\right)$

$H(D) = -\left[\frac{1}{2}\lg\left(\frac{1}{2}\right) + \frac{1}{2}\lg\left(\frac{1}{2}\right) + \frac{1}{2}\lg\left(\frac{1}{2}\right)\right] = -3 \times \frac{1}{2} \times \lg\left(\frac{1}{2}\right) = -\frac{3}{2} \cdot (-1) = \frac{3}{2}$

Q. For a dataset with k classes, what's the maximum and minimum entropy?

→ Equal # of classes in every class → Entropy ↑↑

Entropy

max $\lg k$

min 0

$\text{Entropy} = -\sum P_i \lg P_i$

$= -\sum_{i=1}^{k} \frac{1}{k} \lg \frac{1}{k}$

$= -k \cdot \frac{1}{k} \lg \frac{1}{k}$

$= \lg k$ ← max entropy

All instances in same class → Entropy ↓↓

For 1 particular P_i, $P_i = 1$. Rest, $P_i = 0$

$\text{Entropy} = -[1 \lg 1 + 0 + 0 + \cdots + 0] = 0$ ← min entropy

Information gain

- It quantifies the reduction in entropy after a dataset is split on a particular feature.

$$\mathrm{IG}(D, A) = \mathrm{H}(D) - \sum_{v \in \mathrm{Values}(A)} \frac{|D_v|}{|D|} H(D_v)$$

← weighted sum of entropies

D → Dataset D_v → subset with value v.
A → Feature on which we are splitting
|D| → # of features

Eg: Consider a dataset with following class distribution:-

Class	# instances
A	10.
B	5.
C	5.

Assume that a split on a certain feature F with two values in the dataset results in two datasets with following composition:-

High :- {A:5, B:3, C:2}
Low :- {A:5, B:2, C:3}

Calculate the information gain if we split the dataset on feature F.

→ $IG(D, F) = H(D) - \sum_{v \in \text{Values}(F)} \frac{|D_v|}{|D|} \cdot H(D_v)$

$H(D) = -\sum P_i \lg(P_i) = -\left[\frac{1}{2}\lg\frac{1}{2} + \frac{1}{4}\lg\frac{1}{4} + \frac{1}{4}\lg\frac{1}{4}\right] = \frac{3}{2} = 1.5$

$H(D_{high}) = -\sum P_i \lg P_i = -\left[\frac{5}{10}\lg\frac{5}{10} + \frac{3}{10}\lg\frac{3}{10} + \frac{2}{10}\lg\frac{2}{10}\right] = 1.4855$

$H(D_{low}) = -\sum P_i \lg P_i = -\left[\frac{5}{10}\lg\frac{5}{10} + \frac{2}{10}\lg\frac{2}{10} + \frac{3}{10}\lg\frac{3}{10}\right] = 1.4855$

$|D_{high}| = 10 \qquad |D| = 20$

$|D_{low}| = 10$

$IG(D, F) = 1.5 - \left[\frac{10}{20}(1.4855) + \frac{10}{20}(1.4855)\right] = 1.5 - 1.4855 = 0.0145$

IG is really low.

Choosing the best split:
- The best feature for splitting is the one that provides the highest information gain or the lowest Gini index after the split.

Q: Consider a dataset containing 8 samples with the following class distribution:-

Class	Count
A	5
B	3

Calculate (i) Gini Index of the dataset
(ii) Entropy of the dataset

→ $H(D) = -\left[\frac{5}{8}\lg\left(\frac{5}{8}\right) + \frac{3}{8}\lg\left(\frac{3}{8}\right)\right] = 0.9544$

$GI = 1 - \sum P_i^2 = 1 - \left(\frac{5}{8}\right)^2 - \left(\frac{3}{8}\right)^2 = 0.4687$

Q.62 Details of ten international cricket games between two teams "Green" and "Blue" are given in Table C. This table consists of matches played on different pitches, across formats along with their winners. The attribute *Pitch* can take one of two values: spin-friendly (represented as *S*) or pace-friendly (represented as *F*). The attribute *Format* can take one of two values: one-day match (represented as *O*) or test match (represented as *T*).

A cricket organization would like to use the information given in Table C to develop a decision-tree model to predict outcomes of future games between these two teams.

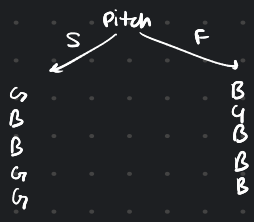
To develop such a model, the computed InformationGain(C, *Pitch*) with respect to the Target is _____ (rounded off to **two** decimal places).

Table C

Match Number	Pitch	Format	Winner (Target)
1	*S*	*T*	Green
2	*S*	*T*	Blue
3	*F*	*O*	Blue
4	*S*	*O*	Blue
5	*F*	*T*	Green
6	*F*	*O*	Blue
7	*S*	*O*	Green
8	*F*	*T*	Blue
9	*F*	*O*	Blue
10	*S*	*O*	Green

$IG(C, Pitch) = H(C) - \sum_{v \in Pitch\ values} \frac{|C_v|}{|C|} \cdot H(C_v)$

$H(C) = -\sum P_i \lg P_i = -\left[\frac{4}{10}\lg\frac{4}{10} + \frac{6}{10}\lg\frac{6}{10}\right] = 0.9709$



$H(C_{Pitch=S}) = -\left[\frac{2}{5}\lg\frac{2}{5} + \frac{3}{5}\lg\frac{3}{5}\right] = 0.9709$ $\quad \frac{|C_S|}{|C|} = \frac{5}{10}$

$H(C_{Pitch=F}) = -\left[\frac{4}{5}\lg\frac{4}{5} + \frac{1}{5}\lg\frac{1}{5}\right] = 0.7219$

$IG(C, Pitch) = 0.9709 - \left[\frac{1}{2}\cdot(0.9709) + \frac{1}{2}(0.7219)\right] = 0.9709 - \frac{1}{2}(1.6928) = 0.9709 - 0.8464$

$= 0.1245$

Constructing decision trees: top-down approach

1. Start with the root node: begin with the entire dataset as the root.
2. Select the best feature: evaluate all possible features using splitting criteria to obtain ... called the decision node.
3. Split the data: Divide the dataset into subsets based on the selected features' values. Each subset will be represented by the branch of the tree.
4. Repeat for each subset: For each subset, repeat the steps 2 and 3.
   - If the subset is homogeneous (all instances belong to the same class), then create a leaf node and assign the class label.
   - If a stopping criterion is met (e.g. maximum depth, exhaustion of features, min # of samples in leaf), create a leaf node with the majority class label.
5. Continue until complete and use the decision tree to assign labels to unknown samples.

Stopping criteria

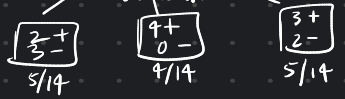
- Maximum depth: Limits the height of the tree to prevent complexity.
- Minimum samples per leaf: Ensures that the leaf nodes contain a minimum # of samples.
- Minimum information gain (or Gini index): Stops splitting if further splits do not significantly improve classification.
- Maximum # of leaf nodes: Limits the total # of leaf nodes (not really good).

consider a dataset. construct a DT using the top-down approach with IG as the splitting criterion.

outlook	temp	humidity	windy	play
sunny 36%	mild 43%	2 unique values	true 6 43%	true 0 0%
rainy 36%	hot 29%		false 8 57%	false 0 0%
Other (4) 29%	Other (4) 29%			
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no

$$H(D) = -\left[\frac{9}{14}\lg\left(\frac{9}{14}\right) + \frac{5}{14}\lg\left(\frac{5}{14}\right)\right] = 0.9403$$

splitting on outlook

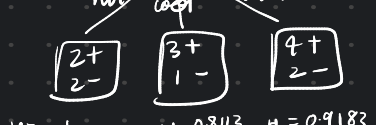


$$IG(D, \text{Outlook}) = H(D) - \left[\frac{5}{14}\times\left[\frac{2}{5}\lg\frac{2}{5} + \frac{3}{5}\lg\frac{3}{5}\right] + \frac{4}{14}\left[\frac{4}{4}\lg\frac{4}{4} + \frac{0}{4}\lg\frac{0}{4}\right] + \frac{5}{14}\left[\frac{3}{5}\lg\frac{3}{5} + \frac{2}{5}\lg\frac{2}{5}\right]\right]$$

$$= 0.9403 - \frac{10}{14}(0.9709)$$

$$= 0.2468$$

splitting on temp:

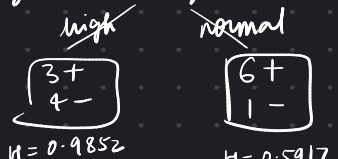


$$IG(D, \text{temp}) = H(D) - \left[\frac{4}{14}(1) + \frac{4}{14}(0.8113) + \frac{6}{14}(0.9183)\right]$$

$$= 0.9403 - 0.9111$$

$$= 0.0292$$

splitting on humidity



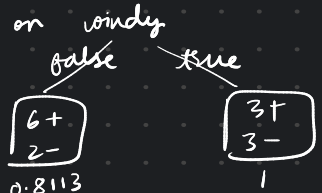
$$IG(D, \text{Humidity}) = H(D) - \left[\frac{7}{14}(0.9852) + \frac{7}{14}(0.5917)\right]$$

$$= 0.9403 - \frac{1}{2}(1.5769)$$

$$= 0.9403 - 0.78845$$

$$= 0.15185$$

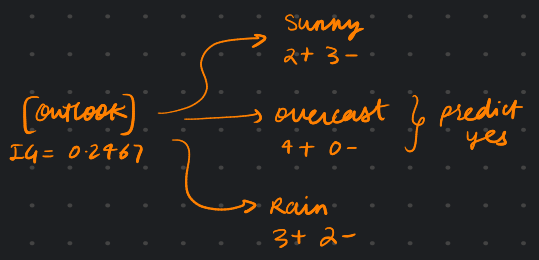
splitting on windy



$$IG(D, \text{windy}) = 0.9403 - \left[\frac{8}{14}(0.8113) + \frac{6}{14}(1)\right]$$

$$= 0.0481$$

outlook gives highest IG.



Now check for sunny:
Dataset left

outlook	temp	humidity	windy	play
sunny	hot	high	false	no
sunny	hot	high	true	no
sunny	mild	high	false	no
sunny	cool	normal	false	yes
sunny	mild	normal	true	yes

$H(D_{sunny}) = 0.971$

splitting on temp
~~hot~~ 0+2- H=0
~~mild~~ 1+1- H=1
~~cool~~ 1+0- H=0

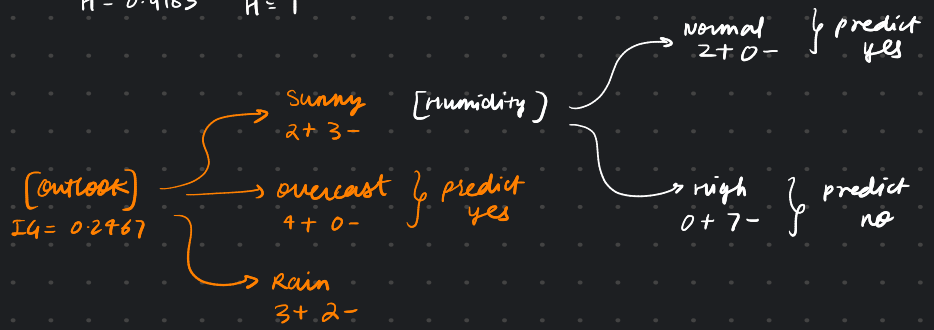
$IG(D_{sunny}, temp) = 0.971 - \left[\frac{2}{5}(0) + \frac{2}{5}(1) + \frac{1}{5}(0)\right] = 0.571$

splitting on humidity
~~normal~~ 2+0- H=0
~~high~~ 0+3- H=0

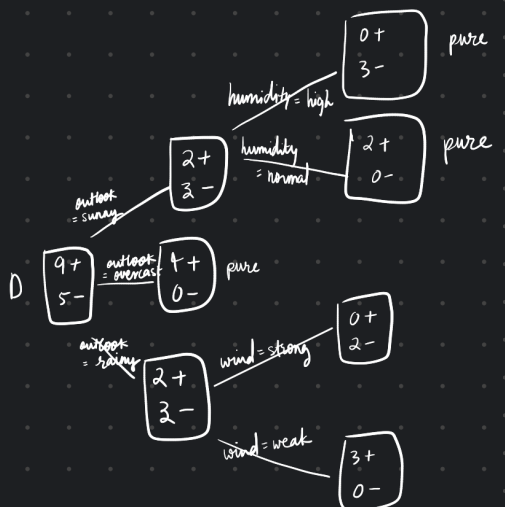
$IG(D_{sunny}, humidity) = 0.971 - 0 = 0.971$ } max

Splitting on windy
~~false~~ 1+2- H=0.9183
~~true~~ 1+1- H=1

IG(D, windy) =

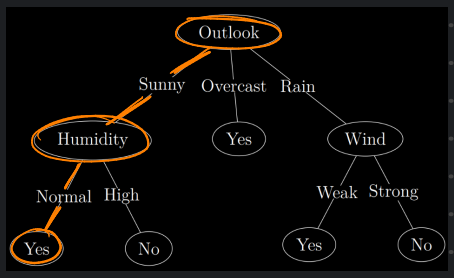


Hy splitting rain on windy gives the best IG.



Q: Classify a new point (outlook = sunny, temp = mild, humidity = normal, windy = false)

→ yes



Handling numerical data

1. Identify unique values: Identify all unique values within the numerical feature in the dataset.
2. Calculate potential splits: For each unique value, consider it as a potential threshold for splitting the data into 2 subsets.
3. Calculate information gain: For each potential split, calculate the IG to tell how much uncertainty is reduced when the dataset is split at that point.
4. Choose the best split: Select the point with the maximum IG.

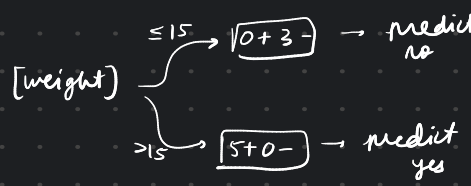
↙ feature ↙ label (mammal or not)

Animal ID	Weight (kg)	Class
1	22	Yes
2	25	Yes
3	47	Yes
4	15	No
5	40	Yes
6	10	No
7	5	No
8	35	Yes

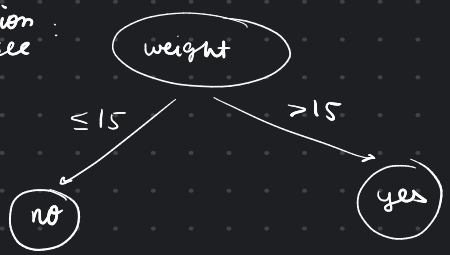
$$H(D) = -\left[\frac{5}{8}\lg\frac{5}{8} + \frac{3}{8}\lg\frac{3}{8}\right] = 0.954$$

Threshold values:	≤ threshold	> threshold	Entropy (weighted)	IG = H(D) − weighted H(attr)
weight: 5	0+ 1−	5+ 2−	0.863	0.0901
10	0+ 2−	5+ 1−		
15	0+ 3−	5+ 0−	0	↓ lowest → IG highest 0.954
22	1+ 3−	4+ 0−		
25	2+ 3−	3+ 0−		
35	3+ 3−	2+ 0−		
40	4+ 3−	1+ 0−		

∴ we'll split on weight ≤ 15, weight > 15



Decision tree:



CART: Classification and regression tree

- Splitting criterion for classification task: Gini index or entropy
  regression task: MSE

Animal ID	Color (4)	Size (3)	Type (3)	Class (2)
1	Brown	Small	Mammal	Yes
2	Black	Small	Mammal	Yes
3	White	Large	Mammal	Yes
4	Black	Large	Reptile	No
5	Green	Small	Reptile	No
6	Brown	Medium	Mammal	Yes
7	Green	Large	Fish	No
8	White	Medium	Mammal	Yes

D: Y:5, N: 3

$Gini(D) = 1-\left[\left(\frac{5}{8}\right)^2+\left(\frac{3}{8}\right)^2\right] \approx 0.468$

Split by Color :

		Gini
Brown	Y:2, N:0	$1-(\frac{2}{2})^2 = 0$
Black	Y:1, N:1	$1-(\frac{1}{2})^2-(\frac{1}{2})^2 = 0.5$
White	Y:2, N:0	0
Green	Y:0, N:2	0

Weighted Gini (Color) $= \frac{2}{8}\times 0+\frac{2}{8}\times 0.5+\frac{2}{8}\times 0+\frac{2}{8}\times 0$

$= 0.125$

Split by Size :

		Gini
Small	Y:2, N:1	0.44
Medium	Y:2, N:0	0
Large	Y:1, N:2	0.44

Weighted Gini (Size) $= \frac{3}{8}\times 0.44+\frac{2}{8}\times 0+\frac{3}{8}\times 0.44$

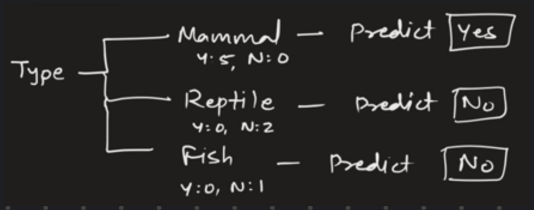
≈ 0.33

Split by Type :

		Gini
Mammal	Y:5, N:0	0
Reptile	Y:0, N:2	0
Fish	Y:0, N:1	0

Weight Gini (Type) = 0

Decision tree :



lower gini ↓
lower entropy ↓

IG = initial gini − Find gini

init ent − Fi- entro

For regression

1. Calculate the mean value: For each leaf node, the predicted value is the mean of the target values of the instances that reach that node.
2. Evaluate Potential Splits: For each feature, determine the potential splits that minimize the MSE.
3. Select the Best split: Choose the split with min. MSE.
4. Repeat Recursively: Continue the process until the stoppi

Eg: Consider a simple dataset containing information about houses and their prices based on two features: Size and Number of Rooms

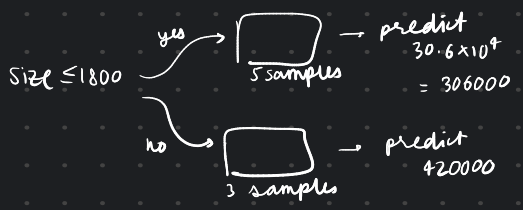
x_1 (numerical) x_2 (categorical) y (continuous)

House ID	Size (sq ft)	Rooms	Price (in $)
1	1500	3	300000 → 30
2	2000	4	400000 → 40
3	2500	4	450000 → 45
4	1800	3	350000 → 35
5	2200	4	410000 → 41
6	1300	2	250000 → 25
7	1700	3	320000 → 32
8	1600	3	310000 → 31

	thresholds	≤ threshold (MSE)	≥ threshold (MSE)	weighted MSE
size:	1300	{25} mean = 25 MSE = $\frac{1\text{st}}{1}[(25-25)^2] = 0$	{30, 40, 45, 35, 41, ~~25~~, 32, 31} mean = 36.3 MSE = 28.49	$\frac{1}{8}(0) + \frac{7}{8}(28.49) = 24.93$
	1500			
	1600			
	1700			
	1800	{30, 35, 25, 32, 31} mean = 30.6 MSE = 10.64	{40, 45, 41} mean = 42 MSE = 4.67	$\frac{5}{8}(10.64) + \frac{3}{8}(4.67) = 8.4$ } lesser MSE, choose this.
	2000			
	2200			

NOTE: check for # of rooms as well.

Say $MSE(D_{size} \le 1800)$ is minimum.



For prediction, just predict the mean of the data points.

Pruning

Overfitting happens when a model is too complex, capturing the noise in the training data rather than the underlying patterns.
Pruning simplifies the model by removing parts of the tree that do not provide significant predictive power.

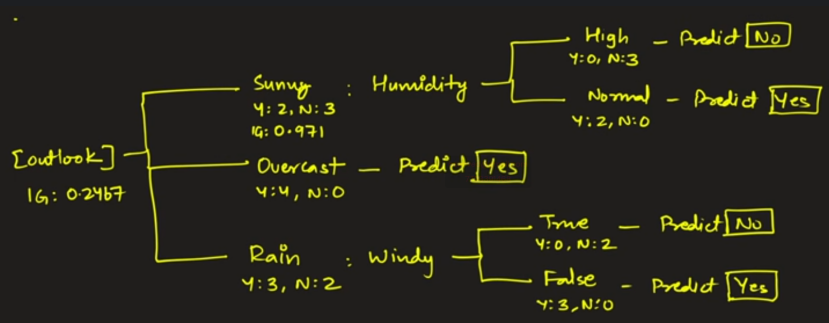
- Pruning
  - pre-pruning (early stopping criteria)
  - post-pruning
    - cost complexity pruning (CCP)
    - reduced error pruning
    - minimum impurity decrease
    - min. leaf size

Cost complexity pruning (weakest link pruning)

- Balance the trade-off between the tree's complexity and its performance on training data.

$$R_\alpha(T) = R(T) + \alpha \cdot |T|$$

$R(T)$: total misclassifica^n rate (cost) of the tree.
$|T|$: # of leaf nodes in the tree
α: non-negative parameter that penalizes the # of terminal nodes
$R_\alpha(T)$: updated mis-classifica^n rate



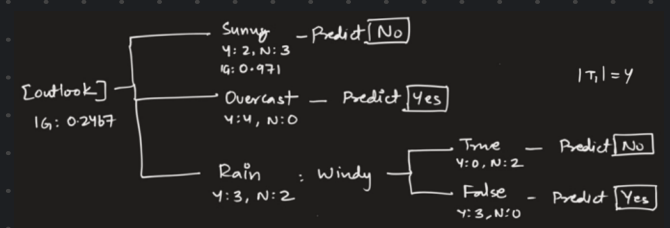
outlook	temp	humidity	windy	play
sunny 36%	mild 43%	2 unique values	true 6 43%	true 0 0%
rainy 36%	hot 29%		false 8 57%	false 0 0%
Other (4) 29%	Other (4) 29%			
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no

$\alpha = 0$:

$R_{\alpha=0}(T) = 0$ → there's no misclassifica^n for the training data (here we stopped after total homogenity).

$\alpha = 0.5$:

$R_\alpha(T) = \frac{0}{14} + (0.5)(5) = 2.5$ } too large

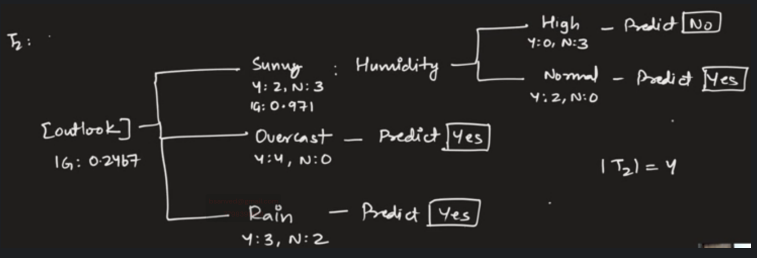


$|T_1| = 4$

$R(T_1) = \frac{2}{14}$

$R_\alpha(T_1) = \frac{2}{14} + (0.5)4 = 2.14$

T_2:

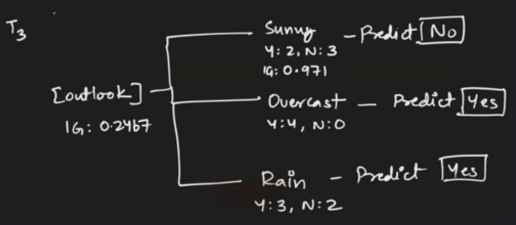


$|T_2| = 4$

$R(T_2) = \frac{2}{14}$

$R_\alpha(T_2) = \frac{2}{14} + 0.5 \cdot 4 = 2.14$

T_3



$|T_3| = 3$

$R(T_3) = \frac{4}{14}$

$R_\alpha(T_3) = \frac{4}{14} + 0.5 \cdot 3 = 1.78$

T_4: Predict – Yes

$|T_4| = 1$

$R(T_4) = \frac{5}{14}$

$R_\alpha(T_4) = \frac{5}{14} + 0.5 \cdot 1 = 0.86$ ← min

$\alpha = 0.5$ gives us T_4 → underfit. take some smaller α.

for $\alpha = 0.1$:

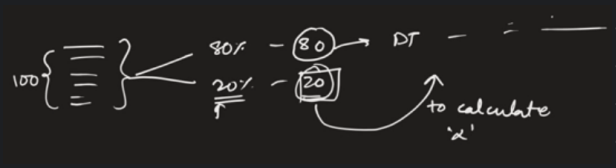
$R_\alpha(T_1) = \frac{0}{14} + 0.1(4) = 0.5$ $\quad R_\alpha(T_3) = 0.58$ $\quad$ underfitting

$R_\alpha(T_2) = \frac{2}{14} + 0.1(4) = 0.54$ $\quad R_\alpha(T_4) = 0.457$ ← min

for $\alpha = 0.02$

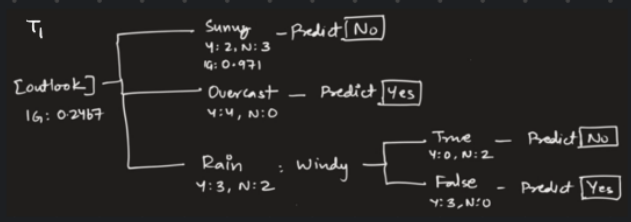
$R_\alpha(T_1) = 0.1$ ← min $\quad R_\alpha(T_2) = 0.22$ $\quad R_\alpha(T_3) = 0.22$ $\quad R_\alpha(T_4) = 0.345$ $\quad R_\alpha(T_5) = 0.37$

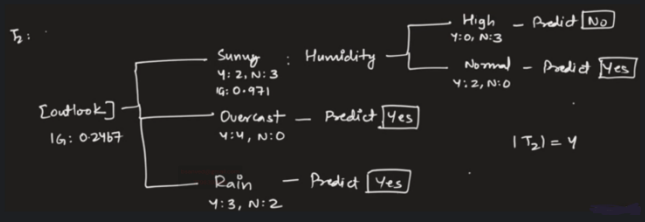
How to select α?
We vary the value of α and calculate the misclassification rate on a separate validation set which is not used in training and choose the α that gives the best results on unseen data.

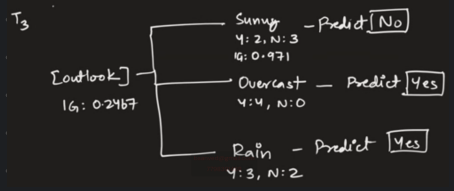


Reduced error pruning

- Start with a fully grown tree.
- For each node, consider removing it and replacing it with a leaf node that predicts the majority class of the training instances that reach the node.
- Measure the error on the validation set for each constructed tree.
- Pick the tree with the min. error on the validation set.







T4 ; Predict – Yes

Validation Set:

Outlook	Temperature	Humidity	Windy	Play Tennis
Sunny	Mild	High	False	No
Overcast	Hot	High	True	Yes
Rain	Cool	Normal	False	Yes
Sunny	Hot	Normal	True	Yes
Rain	Mild	High	True	Yes
Overcast	Mild	High	False	Yes

T_0 : 1/6
T_1 : 2/6
T_2 : 0/6 } min. error tree
T_3 : 1/6
T_4 : 1/6

pick T_2 since it gives the min. error.